

Python Assignment – 3

Question 1 :

What is a user defined function? Write a function to accept two numbers and return their multiplication.

Answer :

```
#Author : codingwithamar@gmail.com
#Description : 'A3Q1 : Write a function to accept two numbers and return
their multiplication'

def Multiplication(No1, No2):
    Ans = 0
    Ans = No1 * No2
    return Ans

def main():
    print("Enter First Number")
    Value1 = int(input())

    print("Enter Second Number")
    Value2 = int(input())

    Ret = Multiplication(Value1, Value2)

    print("Your Multiplication is :", Ret)

if __name__ == "__main__":
    main()
```

Question 2 :

What is the difference between:

- Function with parameters
- Function without parameters Give one example of each.

Answer :

```
#Author : codingwithamar@gmail.com
#Description : 'A3Q2 : Function Parameter'

def display():
    print("Welcome Amar – Without Parameter")
```

```
def Display(name):
    print("Welcome ", name)

def main():
    display()
    Display("AMAR")

if __name__ == "__main__":
    main()

#OUTPUT :
#Welcome Amar - Without Parameter
#Welcome  AMAR
```

Question 3:

Predict the output:

```
def fun():
    x = 10
    print(x)
```

```
fun()
print(x)
```

Explain the reason

Answer :

```
def fun():          #function define, Allocate but not execute in first
    x = 10          #local variable
    print(x)        #print 10 and scope ended and function control return

fun()               #now called function goto control line no - 1
print(x)            #now try to print but "x" not found because x define
inside local function due to scope ended he show error

"""
Code Flow - line 5 > 1 > 2(assign) > 3(print value of x) > 5(function
return) > 6(try to print but cannot) > then show error
OUTPUT :
10
Traceback (most recent call last):
  File "C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-
P229\Assignment-03\A3Q3.py", line 6, in <module>
    print(x)
    ^
NameError: name 'x' is not defined
"""
```

```

"""
Explanation :
Local variables can be accessed only inside the function in which they are
created.

code correction -
def fun():
    x = 10
    return x

x = fun()

print(x)
"""

```

Question 4 :

Write a function that does not return anything but prints a message. Explain the default return value of such a function.

Answer :

A function that does not return anything can still perform actions such as printing messages. If a function does not contain a return statement, Python automatically returns None, which is the default return value of such a function.

```

"""
Write a function that does not return anything but prints a message.
Explain the default return value of such a function.
"""

def display():
    print("Inside Display Function")

result = display()

print("Return Value is : ", result)
print(type(result))

"""
OUTPUT :
Inside Display Function
Return Value is :  None
<class 'NoneType'>
"""

```

Question 5:

What is the difference between: print() and return
Explain with function example.

Answer :

```
def madam(a, b):  
    print(a + b)  
#Display Output - madam ne board var value lihun dakhavali pan aplyala  
dili nahi  
  
def sir(a, b):  
    return a - b  
#send value to caller - siranni ne value dakhavali nahi pan eka kagdavar  
lihun dili  
  
MadamchiBook = madam(10, 20)  
#madam ne value board var dakhavali pan aplyala lihun dili nahi tyamule  
aplya vahit kahich nahi ahe  
SiranchiBook = sir(30, 10)  
#siranni bookmadhe value lihun dili tyamule to kagad aplyakade aahe ani  
tyat value aahe  
  
print("Madamchi Book : ", MadamchiBook)  
#madam fakt dakhavali tyamule aplyakade kahich nahi - none  
print("Siranchi Book : ", SiranchiBook)  
#siranni kagad dilaay tyat value aahe ti apan pahili -  
  
print("Siranchi Value + 1 : ", SiranchiBook + 1)  
#siranchi value reuse keli  
print(MadamchiBook + 1)  
#madamachi valuech nahi tyamule usech nahi honar      #Error
```

Question 6:

Write a program to display:

- Data type
- Memory address
- Size in bytes of a variable entered by the user.

Answer :

```
import sys  
  
def VarRet():  
    var = input("enter the variable : ")  
    return var
```

```
variable = VarRet()
print("User Entered Variable is : ", variable)
print("User Entered Variable Data Type is : ", type(variable))
print("User Entered Variable memory address is : ", id(variable))
print("User Entered Variables Size is : ", sys.getsizeof(variable))
```

Question 7:

Predict the output:

```
a = 5
print(type(a))
```

```
a = 5.5
print(type(a))
```

```
a = "Python"
print(type(a))
```

What Python feature does this demonstrate?

Answer :

```
a = 5
print(type(a))  #"int"
a = 5.5
print(type(a))  #"float"
a = "Python"
print(type(a))  #"str"

a
<class 'int'>
<class 'float'>
<class 'str'>
```

Feature Demonstrated:

This program demonstrates **Dynamic Typing** in Python. A variable is not bound to a specific data type and can refer to objects of different types during execution. The type is associated with the object, not the variable.

Question 8 :

Explain why the following code works without declaration:

`x = 100` Compare this with C or Java.

Answer :

```
X = 100
```

In case of C and Java is the statically type language in there language we declared first data types and other details but Python is the Dynamically typed language and there PVM dynamically allocate Memory space and recognised his data type dynamically at the run time.

In Python, variables do not need explicit declaration before use.

```
x = 100 #python  
  
int x;  
int x = 100; #C and Java
```

Automatically Creates an integer object 100. Automatically Creates the variable name x.
Automatically Makes x refer to the object 100.

Question 9:

What is the difference between:

x = 10

x = "Ten"

Is this allowed? Why?

Answer :

Here 1st assign value is integer and 2nd assign value is str,
It is Allowed. But x value will overwrite and allocate new data id or memory address, because his variable is same but his data type is different. And In python variable not tightly(permanently)d coupled, in case c, java their variable is tightly coupled. And I will not allowed in both language runtime changed datatype.

1st allocate x = 10 with reference count 1 after redeclared x It will overwrite and allocate new memory address.

```
x = 10  
print("In case of int : ", x)  
print(id(x))  
print(type(x))  
  
x = "Ten"  
print("In case of str : ", x)  
print(x)  
print(id(x))  
print(type(x))
```

Question 10 :

Explain how Python manages memory automatically. Why does the programmer not need to explicitly allocate or free memory?

Answer :

Python manages memory automatically using a combination of Memory Allocation, Reference Counting, and Garbage Collection. Whenever an object is created, Python allocates memory for it and keeps track of how many references point to that object. When the reference count becomes zero, the object becomes unreachable and its memory is automatically released by the Garbage Collector. Therefore, the programme

```
import sys  
  
x = [10,20,30]
```

```
print("Address :", id(x))
print("Size      :", sys.getsizeof(x))

y = x

print("Same Address :", id(y))

del x

print(y)
print("Size      :", sys.getsizeof(y))
```